

PERFECT SKILLS

Face Recognition Attendance System

FRAS

Automated CCTV-Based Attendance Capture

150 Classrooms

Scale

7,000 Students

Database

On-Premise

Deployment



What We'll Cover

01

The Problem

Why manual attendance is broken at scale

02

Proposed Solution

FRAS — automated, on-premise, camera-driven

03

Why Perfect Skills

Our track record and credentials

04

What We Need

Server, cameras, and student data

05

How We Do It

End-to-end workflow and technology

06

Timeline & Options

3-week plan and attendance capture modes

07

FAQ

Common questions answered

The Problem

Manual attendance at scale is broken — and it's costing your institution every single day.

5-10 min

lost per class period
to roll-call

150×

classrooms doing
this manually

0%

proxy attendance
prevention

Days

to compile
attendance reports



Time Wasted

Faculty spend 5–10 min per period on roll-call — time that should go to teaching.



Proxy Attendance

Students mark absent peers as present. Academic integrity and UGC compliance are at risk.



No Real-Time Data

Administration can't monitor attendance live — reports are compiled manually, often days later.



Siloed & Error-Prone

Disconnected registers and spreadsheets make audits and student alerts unreliable.



Impossible to Scale

At 7,000 students and 150 rooms, any manual system is fundamentally unscalable.

Proposed Solution

FRAS uses existing CCTV cameras to automatically capture and record attendance — with zero manual effort and zero data leaving your premises.

100%

On-Premise

No cloud. Your data stays on campus.

Zero

Manual Effort

Fully automated attendance capture.

90%

Accuracy

Tested in real classroom conditions.

150

Rooms Simultaneously

One server handles the entire campus.

How FRAS Solves Each Problem



Manual roll-call



Camera captures automatically — no faculty action required



Proxy attendance



Face recognition = physical presence is the only valid proof



No real-time visibility



Live dashboard updated after every session



Siloed data



Centralised DB with Excel/PDF exports and automated alerts

Why Perfect Skills

7

Educational Institutions
Served in last 12 months

3 Weeks

Average Deployment
Time at Scale

90%

Identification Accuracy
in Live Environments

Our Differentiators



On-Premise Expertise

We deploy entirely on your local infrastructure. No cloud, no data privacy concerns.



Proven AI Stack

InsightFace + ArcFace + FAISS — the same stack used by enterprise identity systems.



End-to-End Ownership

Hardware to handover — we own the full project. One point of contact.



Fast, Low-Disruption Rollout

3-week phased deployment designed around academic schedules.

What We Need — Server

All processing happens locally. No internet required for day-to-day operation.

Processor	Intel Xeon or AMD EPYC — 8+ cores recommended
GPU	NVIDIA GPU with 8GB+ VRAM (e.g. RTX 3070, T4, A10) Critical for processing 150 cameras simultaneously. CPU-only will be too slow.
RAM	32 GB minimum — 64 GB recommended
Storage	2 TB SSD for OS, software, and student DB Additional HDD for video buffer if required
Network	Gigabit LAN — cameras and server must be on the same local network
OS	Ubuntu 22.04 LTS (recommended) or Windows Server 2019+

What We Need — Cameras

Camera placement and quality are the single biggest factor in system accuracy.



Resolution	Minimum 1080p Full HD. Existing 2K CCTV can be reused if RTSP-compatible.
Protocol	RTSP support is mandatory for live video ingestion into the processing server.
Mounting Position	Center of classroom, above the blackboard. Front-corner mounting reduces accuracy. Center mount achieves 90–95%.
Mounting Height	7–8 feet from the floor, with 15–20° downward tilt for optimal frontal face capture.
Recommended	Hikvision DS-2CD1023G0E CP Plus CP-UNC-TA21PL3 Any 1080p+ IP cam with RTSP
Lighting	Even lighting across the full seating area. Backlit rooms reduce accuracy by 10–15%. LED panel lighting recommended.

What We Need — Student Data



Photos per Student

3–4 photos minimum. Mix of front-facing and slight-angle shots preferred. College ID card photos are acceptable as the primary photo.



Photo Quality

Clear, well-lit, unmasked. Avoid blurry or heavily filtered images. Good enrollment photos directly improve identification accuracy.



Student Metadata

Student ID, Full Name, Class/Section, Roll Number — provided in Excel or CSV format.

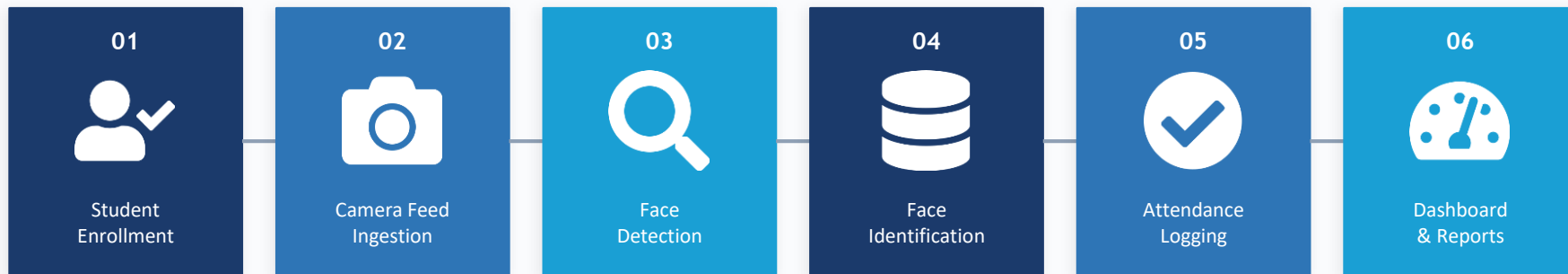


Classroom Mapping

List of all 150 classrooms with the students/sections attending each room — used for automated attendance tagging.

How We Do It — End-to-End Workflow

6-step pipeline — fully automated, entirely on-premise.



01 Student Enrollment

3–4 photos per student → ArcFace generates 512-dimensional embeddings → stored in local vector DB. No photos retained after enrollment.

02 Camera Feed Ingestion

CCTV cameras stream via RTSP to the local server. Frames captured at configurable intervals (hourly / daily / random).

03 Face Detection

InsightFace/RetinaFace scans each frame. Only detections with confidence > 0.75 pass to the next stage.

04 Face Identification

Detected face matched against 7,000-student FAISS vector DB using nearest-neighbour search. Unmatched faces flagged Unknown.

05 Attendance Logging

Student marked Present only when detected reliably across 3+ frames in a session — minimising false positives.

06 Dashboard & Reports

Web dashboard searchable by student, class, date, or subject. Excel/PDF exports. Alerts for attendance < 75%.

Technology Stack & Accuracy

Face Detection

InsightFace (RetinaFace)

State-of-the-art accuracy
for classroom CCTV

Video Ingestion

RTSP Frame Sampling

Frame captured every
2 seconds per camera

Face Recognition

ArcFace 512-D Embeddings

Industry standard for
identification at scale

Backend

Python AI Pipeline

Queue management for
150 simultaneous rooms

Vector Search

FAISS

Sub-millisecond nearest-
neighbour search

Dashboard

Web-Based Portal

Accessible on internal
network — no internet needed

Implementation Timeline — 3 Weeks

WEEK 1

Setup & Enrollment

- Server setup and software installation
- Camera RTSP validation — all 150 classrooms
- Student photo collection and DB enrollment (7,000 students)
- Face embedding generation and vector DB build
- Pilot run in 2–3 classrooms

WEEK 2

Testing & Calibration

- Expand to 15–20 classrooms for accuracy testing
- Measure identification accuracy vs known rosters
- Calibrate confidence thresholds per room
- Handle edge cases — haircuts, glasses, lighting
- Fix misidentifications and tune camera settings

WEEK 3

Rollout & Handover

- Full deployment across all 150 cl
- Dashboard and reporting setup
- Classrooms
- Staff training on attendance portal and exports
- Configure attendance capture mode
- Handover documentation and support SLA

Note: Timeline assumes the server is ready and cameras are RTSP-accessible from Day 1. Hardware procurement should be completed before Week 1.

Attendance Capture Options

Choose one mode — or combine options for maximum effectiveness.

OPTION 1 Every Hour

How

Captures at start of every class period. Attendance logged per period.

Best for

Subject-wise / period-wise attendance (% per subject)

Server load

High — processes 6–8× per day per room

Accuracy risk

Low — multiple data points per day

★ RECOMMENDED

OPTION 2 Once a Day

How

One capture per classroom per day — typically first period.

Best for

Institutions needing a simple daily present/absent record

Server load

Low — one job per classroom per day

Accuracy risk

Medium — one missed capture = full day absent

OPTION 3 Random Interval

How

Captures at unpredictable times within a configured window (e.g. 9am–11am).

Best for

Preventing proxy attendance — students can't predict capture time

Server load

Medium — spread across the day

Accuracy risk

Low — unpredictability deters proxies

★ RECOMMENDED

Our Recommendation

For a college with 150 classrooms and subject-wise attendance requirements — standard for UGC compliance — we recommend:

Option 1

Every Hour — Primary Mode

Ensures subject-wise attendance is captured per period, giving a % attendance figure per subject — required for UGC/university compliance.

Option 3

Random Interval — Add-On

Deters proxy attendance effectively. Students cannot predict when capture happens, making it impossible to cheat.

Option 2

Once a Day — Fallback / Start Option

Can be used if server capacity is a constraint early on, with a clear upgrade path to Option 1 once the system is stable.



We can support a phased rollout — starting with Option 2, then upgrading to Option 1 + 3 as the system stabilises.

Common Questions Answered

? What if a student wears a mask or glasses?

Glasses have minimal impact on ArcFace accuracy. Masks do reduce accuracy — we recommend a mask-free attendance policy during capture windows, consistent with most institutions.

? What happens if the server goes down?

The system queues missed sessions and processes them when the server is back online. Sessions can also be marked manually via the dashboard as a fallback.

? Is student data secure? Who can access it?

All data is stored on-premise — no cloud exposure. Access is controlled via role-based permissions on the dashboard. No photos are retained post-enrollment.

? Can it integrate with our existing ERP / student system?

Yes — we provide attendance data in standard formats (CSV/Excel/API) that can be consumed by most ERP and student information systems. This comes as an Add On

? What accuracy can we realistically expect?

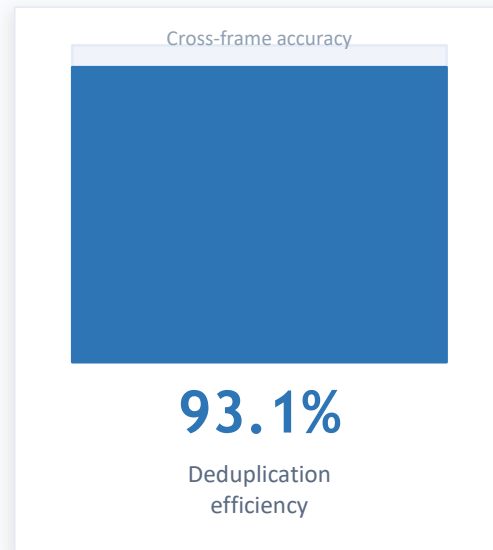
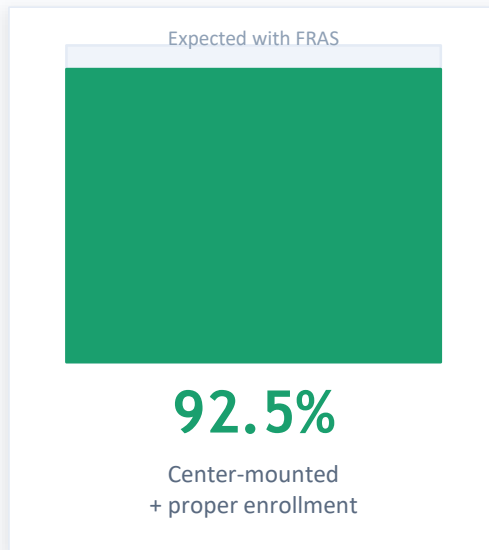
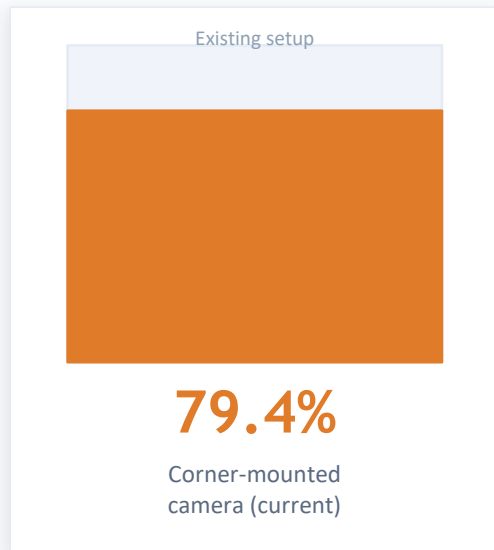
Based on real classroom testing of the shared video: 79.4% with corner-mounted cameras; 90–95% expected with center-mounted cameras and proper enrollment photos.

? What support do we get after go-live?

We provide a defined support SLA as part of the handover based on mutual discussion. This includes issue resolution windows, update support, and a contact escalation matrix.

What to Expect — Accuracy Benchmarks

Based on testing conducted in a real classroom environment — 42 students, 2K CCTV camera.



Key Factors That Drive Accuracy



Camera Position

Center-mount vs corner-mount makes a 12–16% difference in accuracy.



Even Lighting

Backlighting or uneven lighting reduces detection by 10–15%.



Enrollment Quality

3–4 diverse angle photos per student significantly improves matching.



3-Frame Rule

Requiring 3+ frame confirmations minimises false positives dramatically.

What Perfect Skills Delivers



Software Installation

Complete software installation and configuration on the provided server.



Camera Connectivity

RTSP connectivity setup and testing for all 150 cameras.



Student Enrollment

Photo processing, ArcFace embedding generation, and full vector DB build for 7,000 students.



Web Dashboard

Attendance portal deployed on the institution's internal network.



Reports & Alerts

Configured reports, export templates, and attendance threshold alerts.



Staff Training

Hands-on training session for admin and faculty on dashboard usage.



Handover Documentation

Complete system documentation, runbooks, and escalation contacts.



Post-Launch Support

Defined support SLA covering issue resolution and updates after go-live.

Ready to Automate Attendance at Scale?

Let's talk about your campus, your existing cameras, and your timeline.

The logo for perfectskills, featuring the word "perfect" in dark blue and "skills" in orange, both in a lowercase sans-serif font.